

Building intelligent Global Health and Humanitarian
Response Technologies

Week 3: The Last Mile

Context Care at the *fringe*: Community Medicine
Engineering Getting intelligence to the edge

LiGHT

Laboratory for intelligent
Global Health & Humanitarian
Response Technologies.

EPFL

ARIADNE LABS

ASHOKA
UNIVERSITY

**WORLD
ECONOMIC
FORUM**

Rwanda
Centre for the
Fourth Industrial
Revolution

What happens when care leaves the hospital and comes home?

- Who provides care at home, and what knowledge, beliefs, and resources do they draw on?
- What happens to medical advice when it dilutes into community life?
- How can intelligent technologies support care across this last mile?

Care at the *fringe*: Community medicine

Muti, witchcraft and traditional healing

Lecturer: *Prof. Annie Hartley*

Light

Laboratory for intelligent
Global Health & Humanitarian
Response Technologies.

Outline

- 01 Community healers
- 02 Community medicines
- 03 Mechanism of action
- 04 An interview with a witch doctor
- 05 Regulatory framework for traditional medicine
- 06 Western community medicine
- 07 Bonus: A blood sacrifice with a voodoo priest



Traditional healer

Sangoma / Qirha

Southern Africa

Diviner / practitioner specialising in
medicinal remedies

NOT a witchdoctor



Religious healer

Voodoo Priest

Western Africa

Diviner / practitioner specialising in
medicinal remedies

also a witchdoctor

Muti

Muti

Southern Africa

Dawa

East Africa

Jamu...

Indonesia

Medicines and remedies used in southern African healing traditions.

Carry both medicinal & spiritual significance



Types of Muti

- Magical
- Fasting/dieting
- Herbal therapies
- Surgical procedures
- "bleed-cupping"
- Bones of baboons



Modern mixes

Dangers

- Paracetamol
- HIV/TB meds



CONTEXT



NIH Public Access

Author Manuscript

AIDS Behav. Author manuscript; available in PMC 2015 March 01.

Published in final edited form as:

AIDS Behav. 2014 March ; 18(3): 511–518. doi:10.1007/s10461-013-0575-0.

Whoonga: Potential recreational use of HIV antiretroviral medication in South Africa

David J. Grelotti^{1,2}, Elizabeth F. Closson³, Jennifer A. Smit⁴, Zonke Mabude⁴, Lynn T. Matthews^{5,6}, Steven A. Safren^{3,7}, David R. Bangsberg^{5,6,8}, and Matthew J. Mimiaga^{1,3,7}



Routes of administration

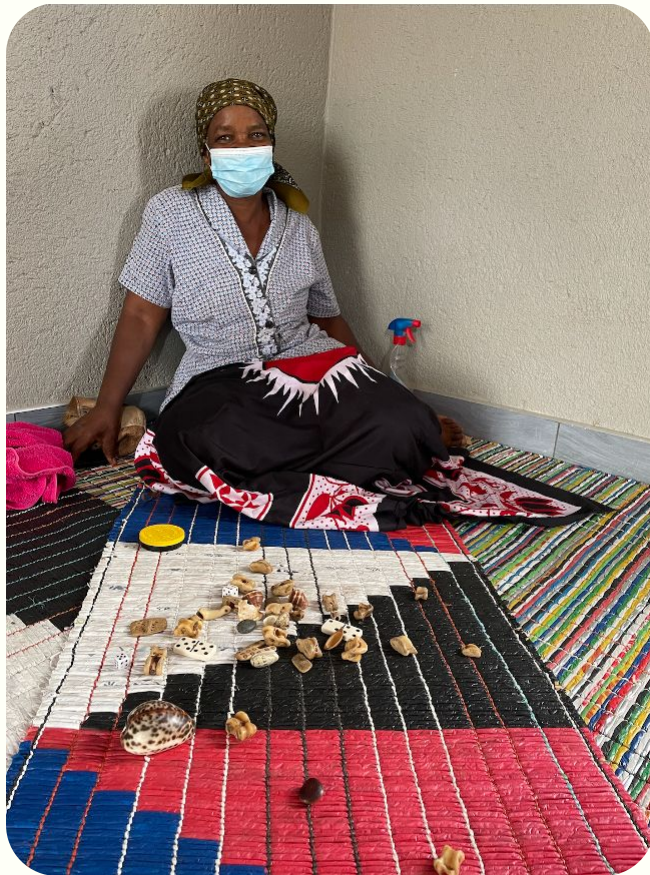
- Bathing
- Vomiting (2 litres lukewarm herbal infusion)
- Steaming
- Nasally (snuff)
- Enemas
- Cuttings (ukugcaba)
- Smoking
- Topical (e.g fat of a boa constrictor)



Access

South Africa (reported, but un certain)

- 200'000 traditional healers (vs 25,000 MDs)
- 60-80% of the population take muti





Belief systems: Perspective from the healer...

An interview with a Sangoma

“The calling”

LiGHT



“My calling”



CONFLICT WITH: “western medicine”

NOT necessarily opposed

- Complementary
- Many shared values
 - Patient centered
 - Killing patients = not good for business
 - Holistic
 - Evidence based
 - Collaborative

But...

- SAFETY
- REGULATION
- STANDARDIZATION
- DOCUMENTATION

LiGHT

Laboratory for intelligent
Global Health & Humanitarian
Response Technologies.





CONFLICT WITH: “western medicine”

An interview with a Sangoma

“White people’s diseases”



Personalized medicine

Illness

- = disagreement with ancestors

Recipe

- = personalized divination

Healer

- = an ordained medium negotiating a truce between ancestry and patient





PHARMACOLOGY: Recipe by divination

An interview with a Sangoma

Consultation
R200 = CHF12

CONTEXT

“Throwing the bones”



LiGHT

Legal Status

Box 1 WHO definition of traditional medicine

‘The sum total of the knowledge, skill and practices based on the theories, beliefs and experiences indigenous to different cultures, whether explicable or not, used in the maintenance of health as well as in the prevention, diagnosis, improvement or treatment of physical and mental illness.’

WHO. Traditional medicine strategy 2014-2023. Geneva: WHO, 2013

Legal Status

Open access

Original research

BMJ Open Legislative landscape for traditional health practitioners in Southern African development community countries: a scoping review

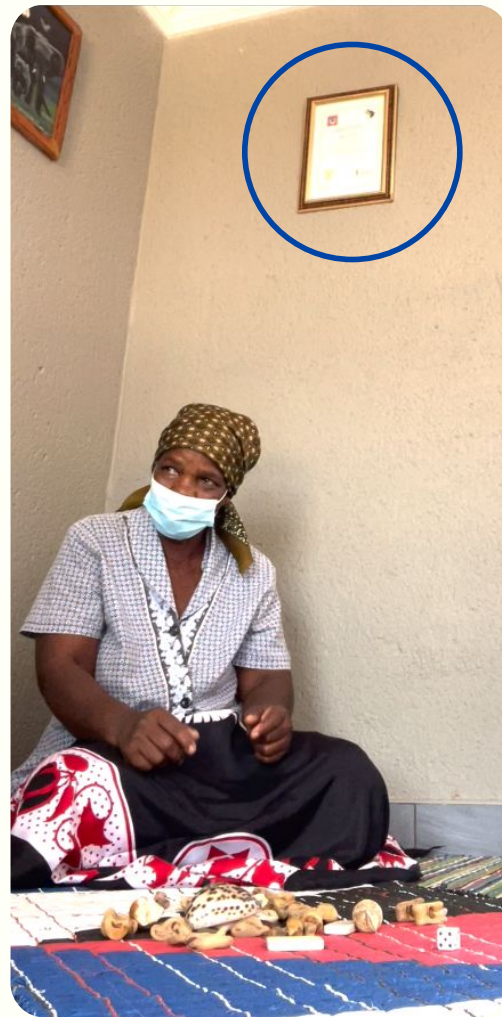
4/14 Southern African = legislation

- South Africa, Namibia and Zimbabwe, Tanzania

3/4 have a council + register to formalise practices

LiGHT

Laboratory for intelligent
Global Health & Humanitarian
Response Technologies.



Certification

Standardised training is **RARE**

- Apprenticeships (observerships)
- Family tradition (oral)
- Government sponsored certification



CONTEXT



RISKS vs BENEFITS: Health education



World Business Markets Breakingviews Video More

HEALTHCARE & PHARMA JUNE 4, 2007 / 9:19 AM / UPDATED 14 YEARS AGO

South Africa's traditional healers help fight HIV

Research article | [Open Access](#) | [Published: 10 June 2019](#)

Traditional healers' role in the detection of active tuberculosis cases in a pastoralist community in Ethiopia: a pilot interventional study

[Bezawit Temesgen Sima](#) , [Tefera Belachew](#), [Gunnar Bjune](#) & [Fekadu Abebe](#)

[BMC Public Health](#) **19**, Article number: 721 (2019) | [Cite this article](#)

1511 Accesses | **3** Citations | [Metrics](#)

LiGHT

Laboratory for intelligent
Global Health & Humanitarian
Response Technologies.

Traditional herbalist

Inyanga

South Africa

Herbalist, pharmacist



Traditional herbalist...at scale



LiGHT

Laboratory for intelligent
Global Health & Humanitarian
Response Technologies.



EPIDEMIOLOGY: Pathological scope

CHRONIC

- Cancer
- Psychiatric disorders
- High blood pressure
- Epilepsy
- Asthma
- Eczema
- HIV/TB
- BPH

ACUTE

- Cholera/Ebola
- Most venereal diseases + UTIs
- Fever

LiGHT

Laboratory for intelligent
Global Health & Humanitarian
Response Technologies.

DR. RASHEED
(A Member of the herbalist Association)
& FORTUNE TELLER
UKUHLOLA YI R40-00

1. Bad luck ie job problems/enemies
2. Help students pass their exams
3. Unsettled Marriages
4. Sexually transmitted diseases (STD,s)
5. How to win gambling ie Lotto, casino, Horses.
6. To chase away tokoloshes
7. Blindness / Body pains / vomiting all the time
8. To stop drinking / smoking
9. We give luck to the poor and the rich
10. We help single people to find partners
11. Diabetes , High / Low Blood Pressure
- 12a. To make your vaginals small or big
- 13b. To restore your virginal elasticity
14. To win loved one, lost love one
15. Make men's penis big & strong
16. People with HIV-AIDS can be helped
16. Body slimming, losing weight with muti
18. Love Attraction
19. Court cases / divorces
20. Men/Women who cannot produce
21. Bewitched people
22. Make more rounds during sex
23. Capable of booming business more customers
24. Short boys to bring money

CELL: 078 4664 662

SONNITEC PRINTERS @ 078 951 7352



A “traditional” AI tool?

Pharmaceutical Equivalence



Medicine unavailable



Photograph a local plant



Identify pharmaceutical equivalents

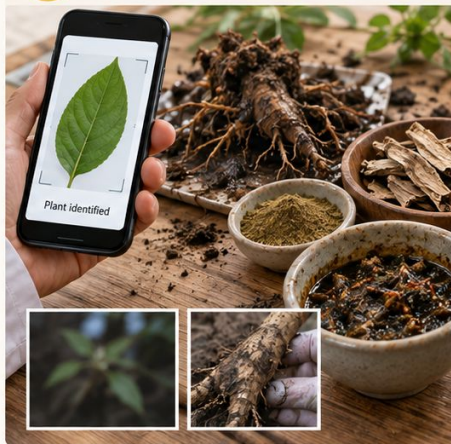


Suggest the plant as a medicine substitute

Proposed by WHO. What is your advice?

A “traditional” AI tool? –why it breaks...

1 INPUT VARIABILITY



2 MISSING EVIDENCE



3 UNSUPPORTED SUBSTITUTION



Proposed by WHO.

LiGHT

Laboratory for intelligent
Global Health & Humanitarian
Response Technologies.

A better “traditional” AI tool?

Compendium of Traditional Knowledge



Engage with traditional healers



Catalogue plants and practices



Healer consensus and standardisation



Scientific review: evidence, gaps and warnings

Community knowledge • Shared standards • Scientific review

Proposed by LiGHT as an alternative

LiGHT

Laboratory for intelligent
Global Health & Humanitarian
Response Technologies.

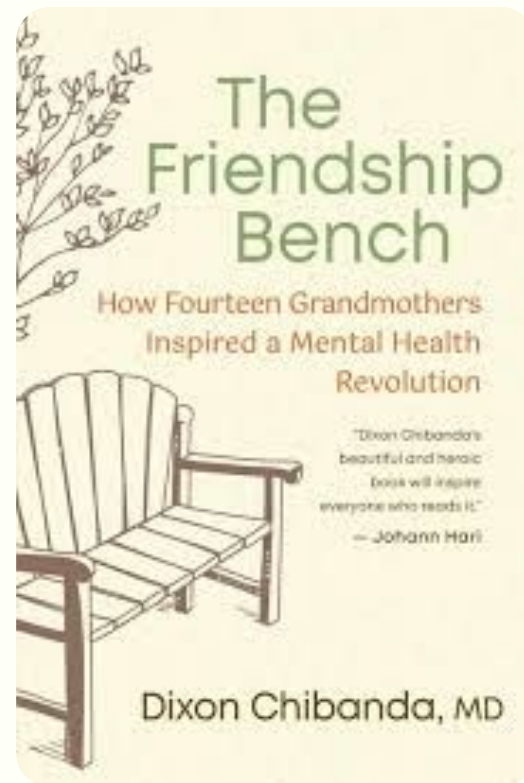
What if we didn't have Traditional medicine?



Community interventions

CONTEXT

ENGINEERING



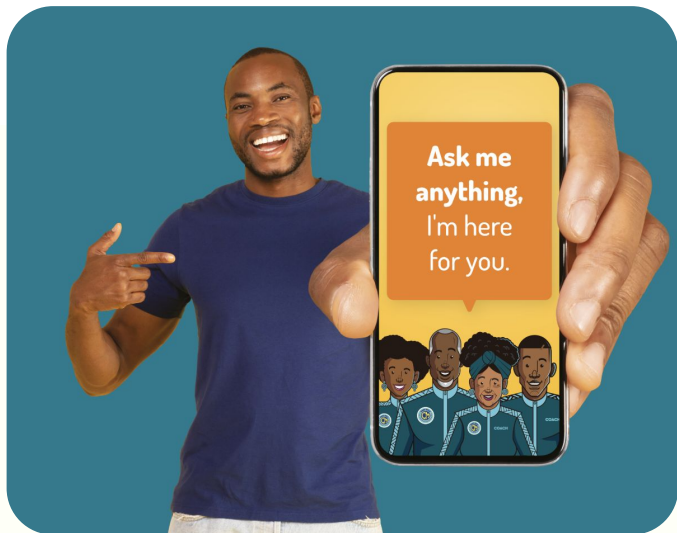
LiGHT

Laboratory for intelligent
Global Health & Humanitarian
Response Technologies.

Community AI interventions

What is the Coach Mpilo AI health coach?

It is a free service that you can use on WhatsApp anytime to get confidential information, advice, and support on relationships, sex, love, family, friends, HIV, TB, and your general health and wellbeing. It is specifically designed to support people living in South Africa.



“I overheard a coach talking to a man and then I understood what coach Mpilo is all about. He can talk to that man in a way I could never do. Men [living with HIV] can communicate with each other like that.”

— Clinic Nurse, Ugu District

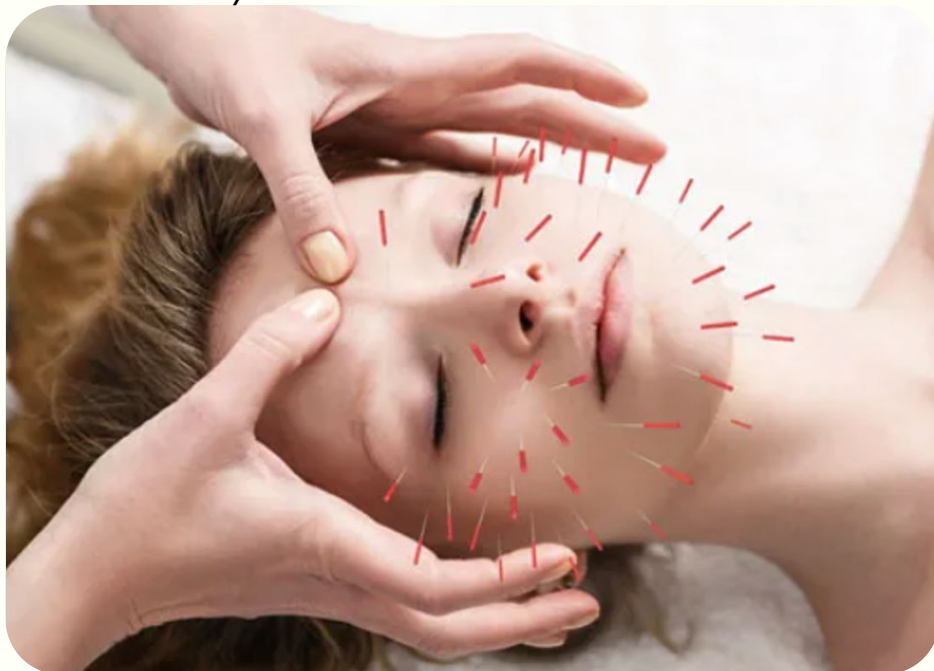
“Western” community medicine

Is it *really* that different?



“Western” community medicine

Is it *really* that different?



LiGHT

Laboratory for intelligent
Global Health & Humanitarian
Response Technologies.

“Western” community medicine

Is it *really* that different?



Otonamaki, Jade eggs & Fish therapy

CONTEXT

“Western” community medicine

Is it *really* that different?



LiGHT Laboratory for intelligent
Global Health & Humanitarian
Response Technologies.

Urine drinking & Perineum sunning

CONTEXT

“Western” community medicine

Is it *really* that different?





Bonus

*A blood sacrifice with a
Voodoo Priest*













RISKS vs BENEFITS: Similarities...?



RISKS vs BENEFITS: Similarities...?



The Last Mile: Getting Intelligence *to the Edge*

Lecturer: *Dr. Lars Klein*

Light

Laboratory for intelligent
Global Health & Humanitarian
Response Technologies.

Outline

- 01 The problem statement
- 02 Listen to the patient
- 03 Think, prepare an answer
- 04 Speak to the patient
- 05 Back to your Projects

Think about it

A patient in a rural low-resource context has a medical question.

What's the best they can do?

Think about it

A patient in a rural low-resource context has a medical question.

What's the best they can do?



Think about it

A patient in a rural low-resource context has a medical question.

What's the best they can do?

What if the regular doctor is busy?
(Ethiopia, [1 physician per 10,000 people](#))



Think about it

A patient in a rural low-resource context has a medical question.

We want to provide healthcare advice that is:

- Accessible
- Contextualized
- Reliable

Think about it

AI as a new driver of healthcare
that actually reaches people



Think about it

AI as a new driver of healthcare
that actually reaches people...

What is wrong here?



This should never be patient-facing!

ChatGPT Health performance in a structured test of triage recommendations [paper](#)

This should never be patient-facing!

ChatGPT Health performance in a structured test of triage recommendations [paper](#)

ChatGPT Health [...] we conducted a structured stress test of triage recommendations using 60 clinician-authored vignettes across 21 clinical domains under 16 factorial conditions, yielding 960 total responses.

This should never be patient-facing!

ChatGPT Health performance in a structured test of triage recommendations [paper](#)

ChatGPT Health [...] we conducted a structured stress test of triage recommendations using 60 clinician-authored vignettes across 21 clinical domains under 16 factorial conditions, yielding 960 total responses.

[...]

Among gold-standard emergencies, the system undertriaged 52% of cases, directing patients with diabetic ketoacidosis or impending respiratory failure to 24–48 h evaluation rather than the emergency department, while correctly triaging classical emergencies such as stroke and anaphylaxis.

This should never be patient-facing!

ChatGPT Health performance in a structured test of triage recommendations [paper](#)

ChatGPT Health [...] we conducted a structured stress test of triage recommendations using 60 clinician-authored vignettes across 21 clinical domains under 16 factorial conditions, yielding 960 total responses.

[...]

Among gold-standard emergencies, the system undertriaged 52% of cases, directing patients with diabetic ketoacidosis or impending respiratory failure to 24–48 h evaluation rather than the emergency department, while correctly triaging classical emergencies such as stroke and anaphylaxis.

[...]

Crisis-intervention messages activated unpredictably across suicidal ideation presentations, occurring more frequently when patients described no specific method than when they did.

Work together with partners!

User-centric design.

Partners describe a use-case, we build tailor-made tools.

Work together with partners!

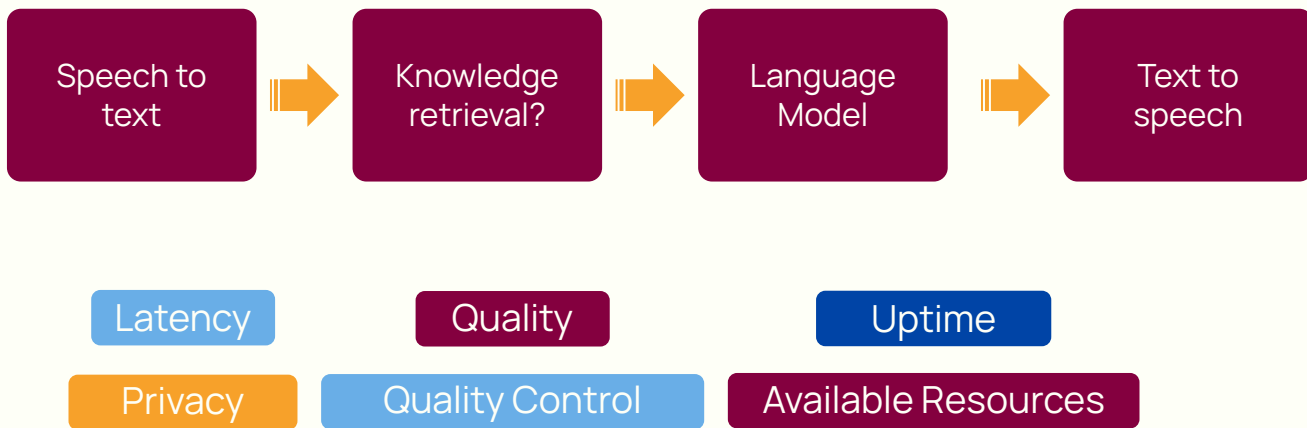
User-centric design.

Partners describe a use-case, we build tailor-made tools.

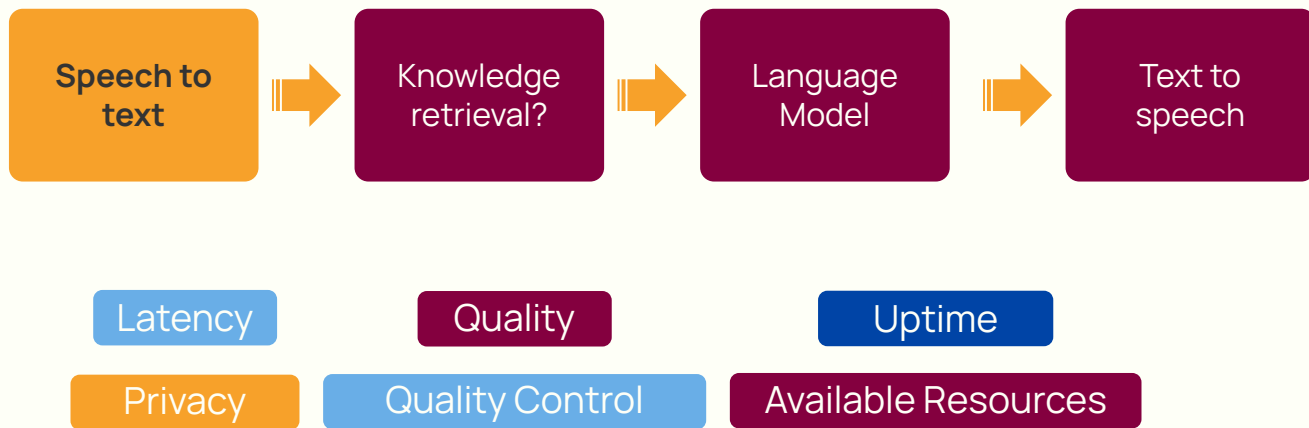
But the challenges are the same:

- Accessible
- Contextualized
- Reliable

End to end flow



01 Listen to the patient



The machinery

The microphone records ~44kHz. Resample to 16kHz

Whisper chunks into 30s and works on log-mel spectrograms.

An encoder reads the spectrogram.

A decoder writes tokens one at a time:

`<|am|>` `<|transcribe|>` then the transcription.

The machinery

The microphone records 16kHz.

Whisper chunks into 30s and works on log-mel spectrograms.

An encoder reads the spectrogram.

A decoder writes tokens one at a time:

`<|am|>` `<|transcribe|>` then the transcription.

Robust Speech Recognition via Large-Scale Weak Supervision

Alec Radford^{*1} Jong Wook Kim^{*1} Tao Xu¹ Greg Brockman¹ Christine McLeavey¹ Ilya Sutskever¹

The machinery

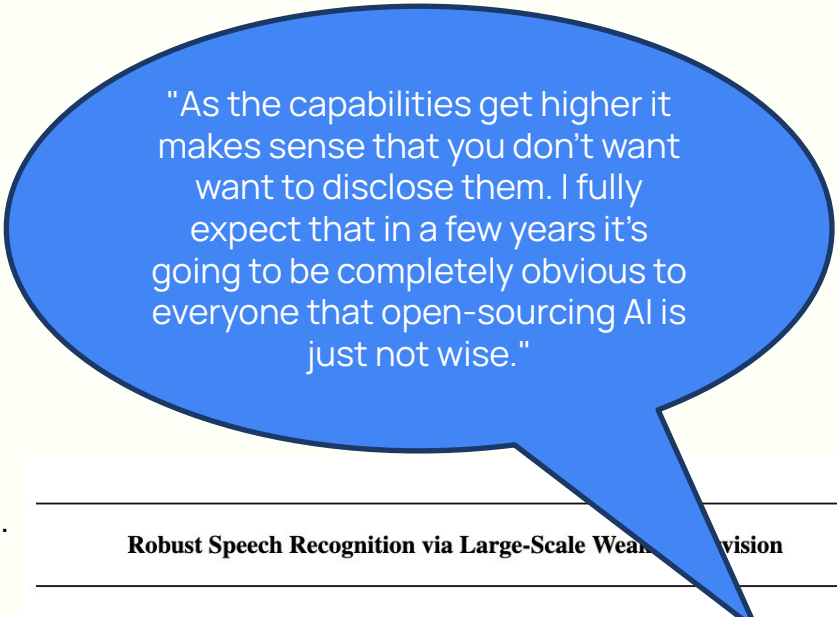
The microphone records 16kHz.

Whisper chunks into 30s and works on log-mel spectrograms.

An encoder reads the spectrogram.

A decoder writes tokens one at a time:

`<|am|>` `<|transcribe|>` then the transcription.



"As the capabilities get higher it makes sense that you don't want to disclose them. I fully expect that in a few years it's going to be completely obvious to everyone that open-sourcing AI is just not wise."

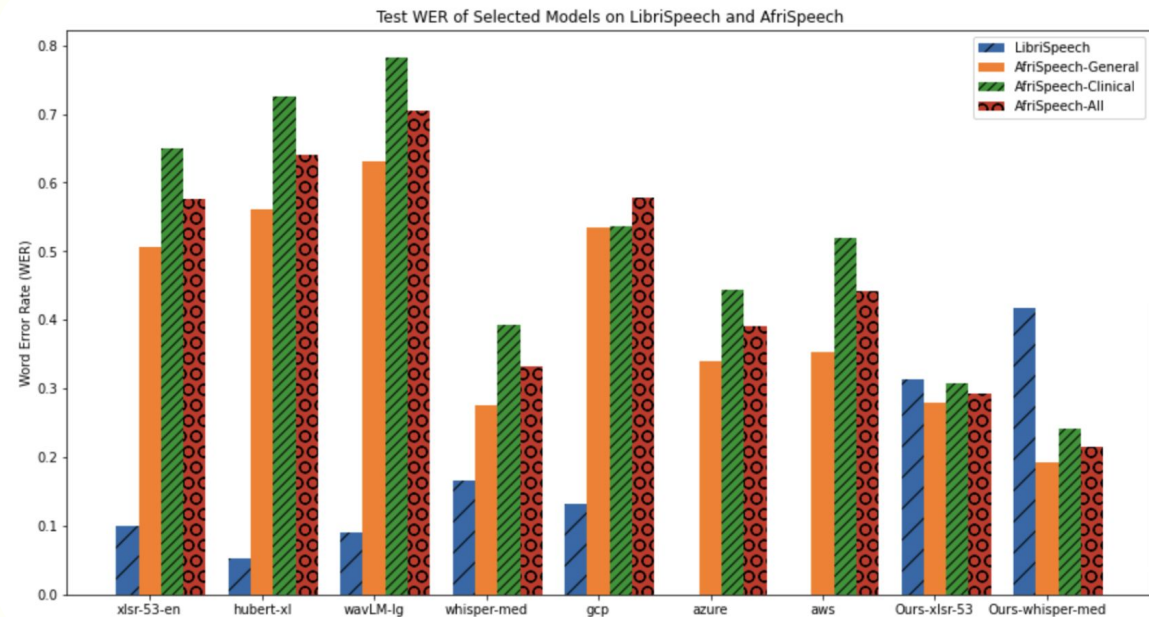
Robust Speech Recognition via Large-Scale Weak Supervision

Alec Radford^{*1} Jong Wook Kim^{*1} Tao Xu¹ Greg Brockman¹ Christine McLeavey¹ Ilya Sutskever¹

Thanks anyways

Model	Parameters	Encoder / decoder layers
tiny	39 M	4 / 4
base	74 M	6 / 6
small	244 M	12 / 12
medium	769 M	24 / 24
large-v3	1.55 B	32 / 32
large-v3-turbo	809 M	32 / 4

Is your data WEIRD?



Beyond whisper

Model	Who	Languages	Size	Note
Whisper large-v3	OpenAI	99	1.55 B	Reasonable default
Moonshine base	Useful Sensors	English + few	61 M	tiny
Parakeet TDT, Canary	NVIDIA	en / 25	0.6 B, 1 B	Top of Open ASR leaderboard
MMS	Meta	1,100+, incl. Amharic	1 B	Look at that language coverage
Voxtral Mini	Mistral	8+	3 B	Multimodal LLM
Kyutai STT	Kyutai	en, fr	1 B, 2.6 B	Agentic?

Ok, now how do you run this?

Engine	Language	Hardware	In the browser?
PyTorch + transformers	Python	CPU/GPU	no
faster-whisper (CTranslate2)	Python	GPU, CPU int8	no
whisper.cpp (GGML)	C++	CPU, Metal, CUDA, phones	yes: WASM, up to small
ONNX Runtime	C++, JS	everything	yes: WASM on the CPU, WebGPU on the GPU
sherpa-onnx	C++	embedded, phones	yes: WASM

Demo time

Ticked quite a few boxes here

Latency

WebGPU acceleration is pretty sweet.

Quality

Depends on the language.

Privacy

Perfect.

Available Resources

A nasty surprise: 2-3x RAM compared to download size.

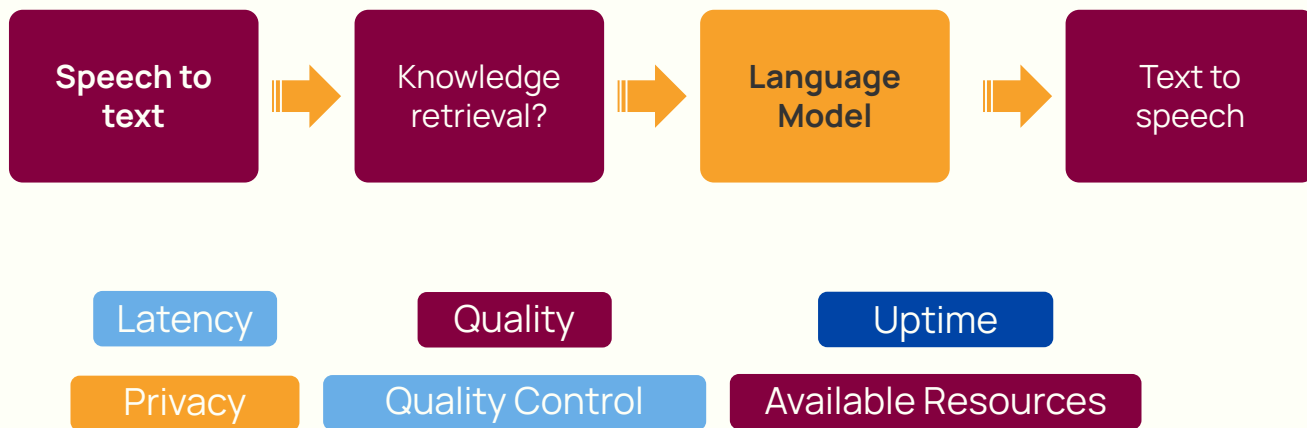
Takeaway messages

Speech to text is basically a solved problem for English.

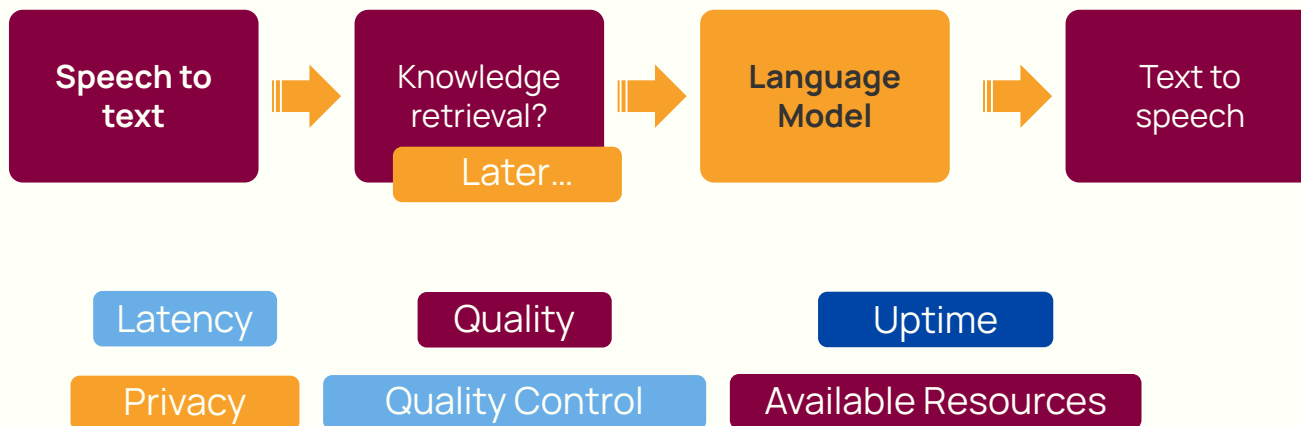
Training data is more important than model size.

Depending on the deployment context, there is no solution at all.

03 Think, prepare an answer



03 Think, prepare an answer

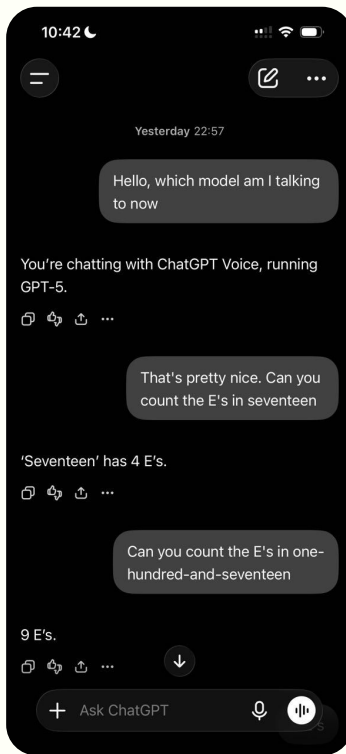


LLMs are everywhere

If you have a hammer, everything looks like a nail.
If you have a reasoning model...

LLMs are everywhere

If you have a hammer, everything looks like a nail.
If you have a reasoning model...



LLMs are ~~everywhere~~

If you have a hammer, everything looks like a nail.

If you have a reasoning model...

What if you have no network?

LLMs are NOT everywhere.

LLMs are ~~everywhere~~

If you have a hammer, everything looks like a nail.

If you have a reasoning model...

What if you have no network?

LLMs are NOT everywhere.

This is where they actually are →



Inside the world's most powerful AI datacenter

LLMs are ~~everywhere~~

If you have a hammer, everything looks like a nail.
If you have a reasoning model...

What if you have no network?
LLMs are NOT everywhere.

Let's talk small large language models.

LLMs are ~~everywhere~~

If you have a hammer, everything looks like a nail.

If you have a reasoning model...

What if you have no network?

LLMs are NOT everywhere.

Let's talk small large language models.

Today, we're releasing **LFM2.5-230M**, our smallest model yet. It's a fast, lightweight foundation for developers to fine-tune and deploy in agentic workflows. Built on the LFM2 architecture, it delivers exceptionally fast inference and runs everywhere, from cloud GPUs to low-cost CPUs (213 tok/s decode speed on Galaxy S25 Ultra, 42 tok/s on a Raspberry Pi 5). Despite its small size, it's surprisingly capable at tool use and data extraction tasks.

Open weights

Family	Who	Sizes	License	Note
Gemma 3, 4	Google	270M, 1B, 4B, 12B, 27B	Gemma terms Apache 2 for Gemma 4	270M runs on anything
Qwen 3.x	Alibaba	0.6B to 235B	Apache 2	strong small models
Llama 3.x, 4.x	Meta	1B, 3B, 8B, 70B	Llama license	used to be the default
Mistral, Ministral	Mistral	3B, 8B, 24B	Apache 2 (mostly)	European
Apertus	Swiss AI	8B, 70B	Apache 2	fully open
MedGemma	Google	4B, 27B	Health AI terms	medical, multimodal

Ok, now how do you run this?

Engine	Language	Hardware	Good for
PyTorch + transformers	Python	GPU	research, fine-tuning
vLLM, SGLang	Python, CUDA	GPU server	many users, one GPU
llama.cpp, Ollama	C++	CPU, Metal, CUDA, phones	one user, one machine
MLX	Python, Swift	Apple silicon	your laptop
ExecuTorch, LiteRT	C++	phones, NPUs	native apps
WebLLM, Transformers.js	JS	browser, WebGPU	lecture demos

What runs where

Where	RAM for the model	Fits	Tokens per second	Needs
Phone	2 to 4 GB	up to 3B, 4 bit	5 to 15	battery, lot of patience
Laptop	8 to 32 GB	up to 27B, 4 bit	10 to 40	
Clinic box, one consumer GPU	~24 GB, larger for unified architectures	up to 27B, or 70B at 4 bit, slowly	30 to 60	
GPU server(s)	80 GB and up	anything	Hundreds (with batching)	The internet 😊
Browser	same as laptop, minus a bit	up to 8B with WebGPU	10 to 30	WebGPU

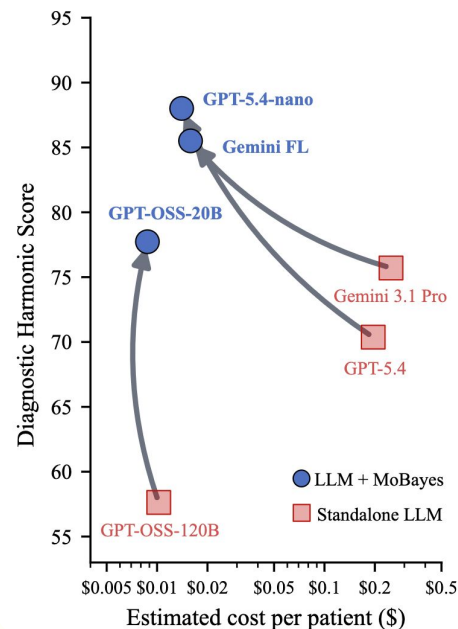
Demo time

Grounding

This isn't even about reasoning.
The first step is memorization.
Small models quickly hit a wall.

RAG is a superpower. That's why it has its own lecture ...

[MOBAYES: A Modular Bayesian Framework for Separating Reasoning from Language in Conversational Clinical Decision Support](#)



So many tradeoffs

Latency

High. You need good networking or good compute.

Quality

Language skills are ok, just don't trust the advice.

Available Resources

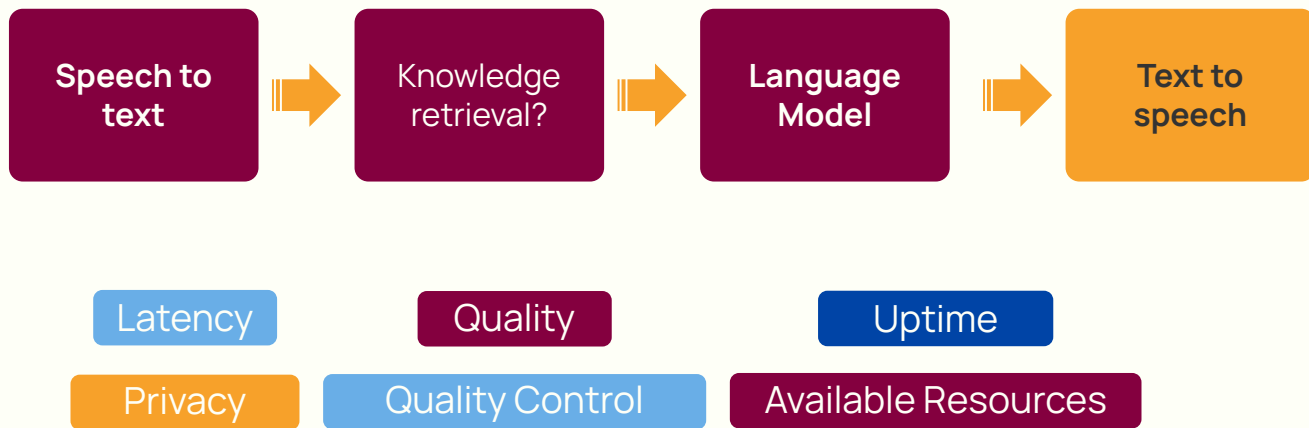
That's the big question.

Takeaway messages

You can get a lot of language out of a small language model.

Small models are fluent and wrong. Bigger models are fluent and less wrong.

04 Speak to the patient



How does this work?

Classic pipeline:

text to phonemes → phonemes to a spectrogram → spectrogram to audio (vocoder)

Kokoro and Piper work like this. Small, fast, one voice per model.

How does this work?

Classic pipeline:

text to phonemes → phonemes to a spectrogram → spectrogram to audio (vocoder)

Small, fast, sometimes just one voice per model.

New pipeline:

LLM that writes audio tokens instead of text tokens.

Bigger, can clone a voice from 3 seconds...

Who can talk

Model	Who	Size	Languages	License
Kokoro	Hexgrad	82 M	English + a few	Apache 2
Pocket TTS	Kyutai	100 M	en, fr, de, es, it, pt	MIT
Piper	OHF-Voice	15 to 60 MB per voice	34, incl. Swahili	GPL
MMS-TTS	Meta	36 M per language	1,100+, incl. Amharic	CC BY-NC
Qwen3-TTS	Alibaba	0.6 B, 1.7 B	10	Apache 2
Voxtral TTS	Mistral	4 B	9	CC -NC
Chatterbox	Resemble AI	0.5 B	20+	MIT

Is your data WEIRD? Part 2

Model	Released	Languages	Amharic	Swahili
MMS-TTS	2023	1,100+	yes	yes
Piper	2023	34	no	yes, one voice
Kokoro	2025	English + a few	no	no
Chatterbox Multilingual	2025	20+	no	no
Qwen3-TTS	2026	10	no	no
Voxtral TTS	2026	9	no	no
Pocket TTS	2026	6	no	no

The newer the model, the shorter the language list.

Ok, now how do you run this?

Engine	Language	Hardware	In the browser?
PyTorch + transformers	Python	GPU, CPU	no
vLLM (Qwen3-TTS)	Python, CUDA	GPU server	no
Piper	C++, Python	CPU, Raspberry Pi	no
sherpa-onnx	C++	embedded, phones	yes: WASM
ONNX Runtime, kokoro-js	JS	everything	yes: WASM on the CPU, WebGPU on the GPU

Demo time

Ticked quite a few boxes here

Latency

Real time on a CPU. Milliseconds to the first audio.

Quality

Again, English is fine ...

Uptime

None of this needs a server.

Privacy

Perfect.

Available Resources

Negligible, the cheapest step in the pipeline.

Takeaway messages

Speech out is the easiest box.

Language coverage is the worst of the three boxes.

The only model that speaks Amharic* is from 2023 and non-commercial.

*please prove me wrong

Questions?

Thank You

Except where otherwise indicated, original content in this presentation is © 2026 Hartley, Keitel, Klein, Sasu, Arni, Kesmen, Nemo and licensed under CC BY 4.0: <https://creativecommons.org/licenses/by/4.0/>.

Third-party materials are excluded from this license and remain subject to their indicated rights.